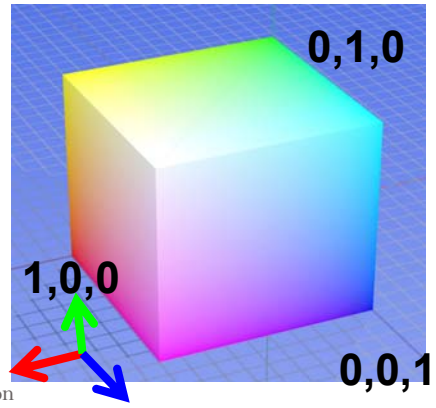
Computer Vision
Lecture 4: Color Transform

Yeungnam University
MSP lab.
Ho-Youl JUNG

영남대학교 Yeungnam University

Primary Color of light

- Default color
- RGB cube
  - Easy for devices
  - But not perceptual
  - Where do the grays live?
  - Where is hue?



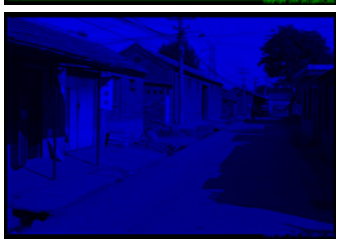




R
(G=0,B=0)



G
(R=0,B=0)



B
(R=0,G=0)

CMY / CMYK representation

- cyan(청록), magenta (자홍) and yellow

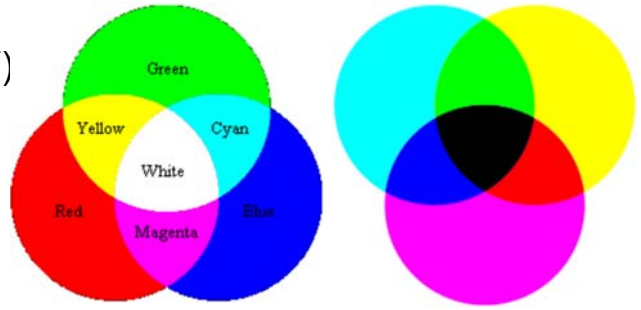
$$C = 1.0 - R$$
$$M = 1.0 - G$$
$$Y = 1.0 - B$$

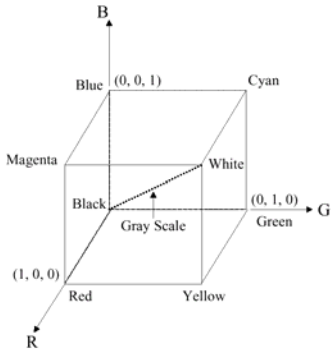
$$\Leftrightarrow$$

$$K = \min(C, M, Y)$$
$$C = C - K$$
$$M = M - K$$
$$Y = Y - K$$

$$\Updownarrow$$

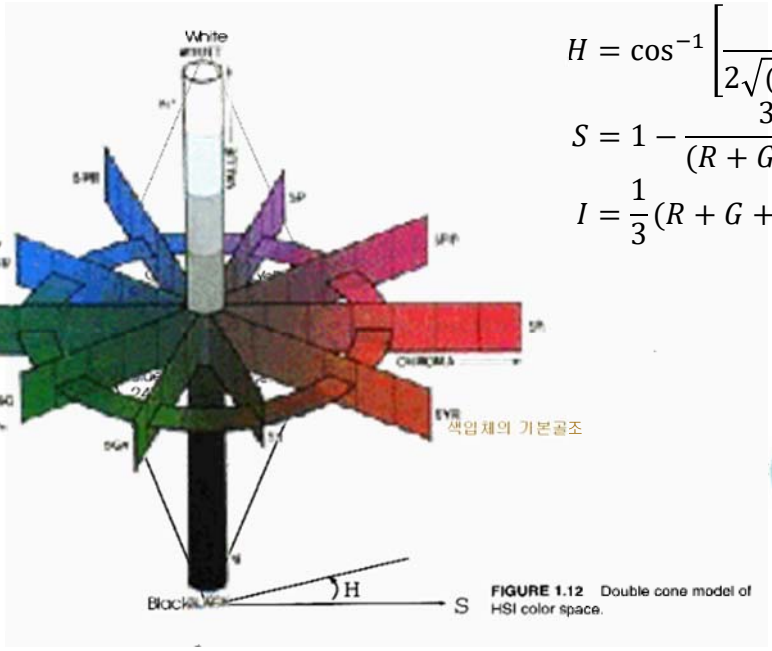
$$R = 1.0 - C$$
$$G = 1.0 - M$$
$$B = 1.0 - Y$$





HSI representation

- 색상(Hue), 채도(Saturation), 명도(Intensity)



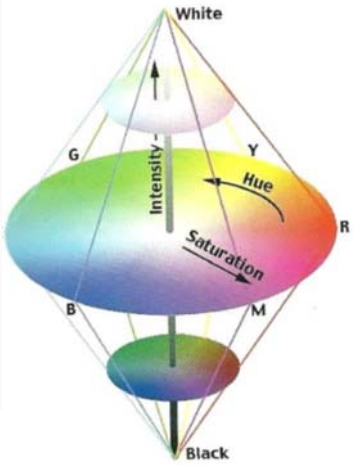
색입체의 기본골조

FIGURE 1.12 Double cone model of HSI color space.

$$H = \cos^{-1}\left[\frac{(R-G)+(R-B)}{2\sqrt{(R-G)^2+(R-B)(G-B)}}\right]$$

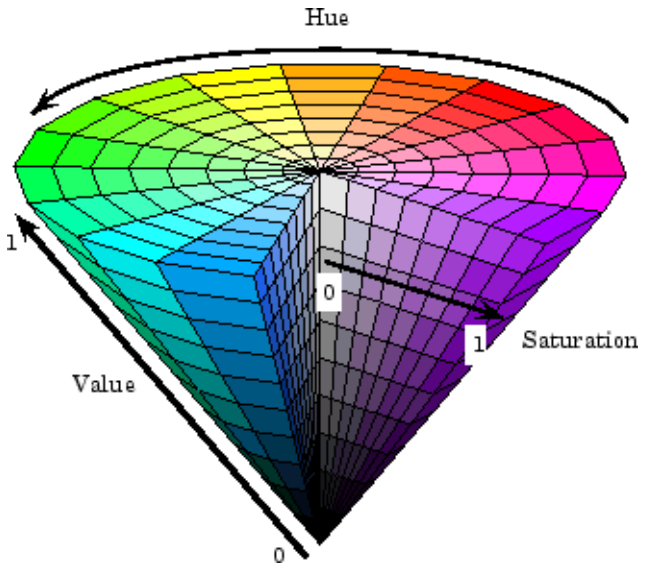
$$S = 1 - \frac{3}{(R+G+B)}[\min(R,G,B)]$$

$$I = \frac{1}{3}(R+G+B)$$

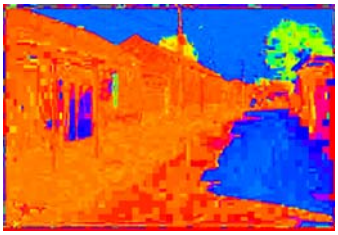


HSI representation

- 색상(Hue), 채도(Saturation), 명도(Intensity)







H
(S=1,V=1)



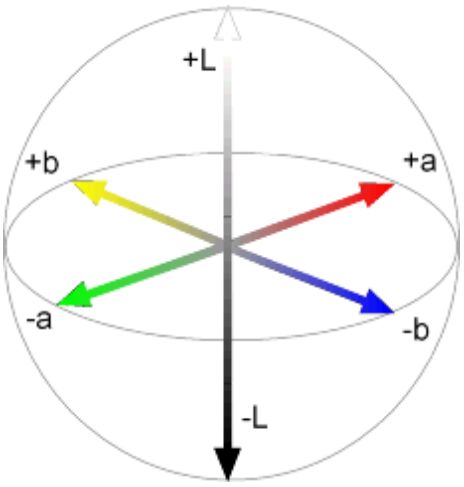
S
(H=1,V=1)



I
(H=1,S=0)

L*a*b*

- Perceptually uniform" color space







L
(a=0,b=0)

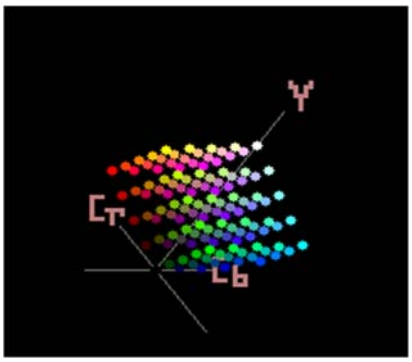


a
(L=65,b=0)



b
(L=65,a=0)

YC_bC_r representation



$$Y = 0.29900R + 0.58700G + 0.11400B$$
$$Cb = -0.16874R - 0.33126G + 0.50000B$$
$$Cr = 0.50000R - 0.41869G - 0.08131B$$

$$A = \begin{bmatrix} 0.2990 & 0.5870 & 0.1140 \\ -0.1687 & -0.3313 & 0.5000 \\ 0.5000 & -0.4187 & -0.0813 \end{bmatrix} \qquad A^{-1} = \begin{bmatrix} 1.0000 & -0.0000 & 1.4020 \\ 1.0000 & -0.3441 & -0.7141 \\ 1.0000 & 1.7720 & 0.0000 \end{bmatrix}$$

$$R = 1.00000Y + 1.40200Cr$$
$$G = 1.00000Y - 0.34414Cb - 0.71414Cr$$
$$B = 1.00000Y + 1.77200Cb$$

YC_bC_r representation

- Digital Image (color): The basic equations to convert between 8-bit digital RGB data with a 16-235 nominal range and YCbCr

➢ For SDTV

$$Y_{601} = 0.299R' + 0.587G' + 0.114B'$$
$$Cb = -0.172R' - 0.339G' + 0.511B' + 128$$
$$Cr = 0.511R' - 0.428G' - 0.083B' + 128$$

$$R' = Y_{601} + 1.371(Cr - 128)$$
$$G' = Y_{601} - 0.698(Cr - 128) - 0.336(Cb - 128)$$
$$B' = Y_{601} + 1.732(Cb - 128)$$

➢ For HDTV

$$Y_{709} = 0.213R' + 0.715G' + 0.072B'$$
$$Cb = -0.117R' - 0.394G' + 0.511B' + 128$$
$$Cr = 0.511R' - 0.464G' - 0.047B' + 128$$

$$R' = Y_{709} + 1.540(Cr - 128)$$
$$G' = Y_{709} - 0.459(Cr - 128) - 0.183(Cb - 128)$$
$$B' = Y_{709} + 1.816(Cb - 128)$$

	Nominal Range	White	Yellow	Cyan	Green	Magenta	Red	Blue	Black
SDTV									
Y	16 to 235	180	162	131	112	84	65	35	16
Cb	16 to 240	128	44	156	72	184	100	212	128
Cr	16 to 240	128	142	44	58	198	212	114	128
HDTV									
Y	16 to 235	180	168	145	133	63	51	28	16
Cb	16 to 240	128	44	147	63	193	109	212	128
Cr	16 to 240	128	136	44	52	204	212	120	128

YC_bC_r representation

- Digital Image (color): If RGB data has a range of 0-255

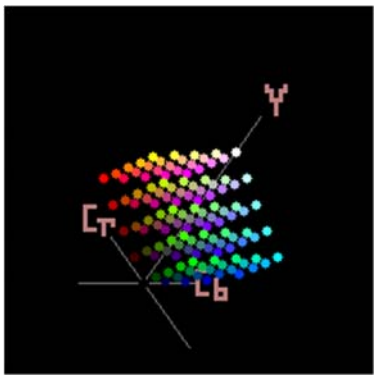
➢ For SDTV: computer system

$$Y_{601} = 0.257R' + 0.504G' + 0.098B' + 16$$
$$Cb = -0.148R' - 0.291G' + 0.439B' + 128$$
$$Cr = 0.439R' - 0.368G' - 0.071B' + 128$$
$$R' = 1.164(Y_{601} - 16) + 1.596(Cr - 128)$$
$$G' = 1.164(Y_{601} - 16) - 0.813(Cr - 128) - 0.391(Cb - 128)$$
$$B' = 1.164(Y_{601} - 16) + 2.018(Cb - 128)$$

➢ For HDTV : Computer system

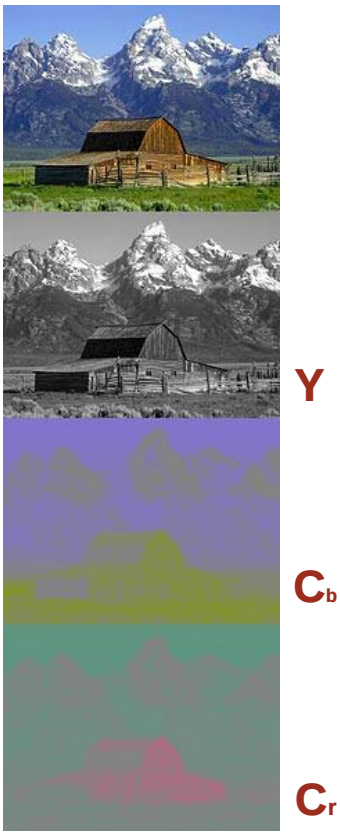
$$Y_{709} = 0.183R' + 0.614G' + 0.062B' + 16$$
$$Cb = -0.101R' - 0.338G' + 0.439B' + 128$$
$$Cr = 0.439R' - 0.399G' - 0.040B' + 128$$
$$R' = 1.164(Y_{709} - 16) + 1.793(Cr - 128)$$
$$G' = 1.164(Y_{709} - 16) - 0.534(Cr - 128) - 0.213(Cb - 128)$$
$$B' = 1.164(Y_{709} - 16) + 2.115(Cb - 128)$$

YC_bC_r representation



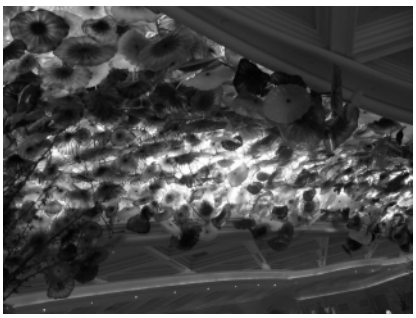
- **C_bC_r Planes at different Y values**



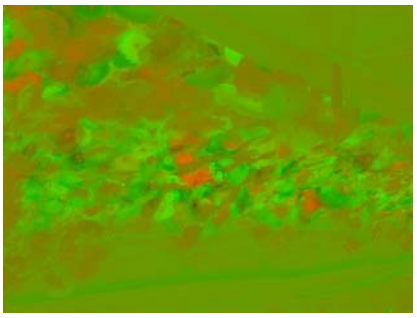


YC_bC_r representation

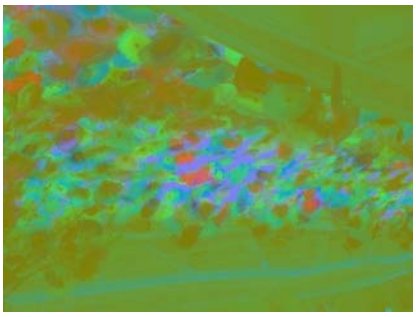




Y



C_b



C_r

YUV representation

- Represented by Luma(Y), and color information(U,V)
- For RGB values with a range 0-255, Y has a range of 0-255, U a range of 0 to+/-112, and V a range of 0 to +/-157

$$Y = 0.299R' + 0.587G' + 0.114B'$$

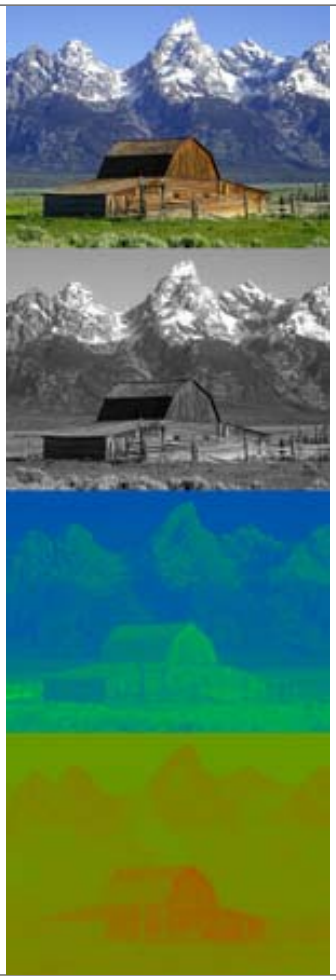
$$U = -0.147R' - 0.289G' + 0.436B'$$
$$= 0.492\,(B' - Y)$$

$$V = 0.615R' - 0.515G' - 0.100B'$$
$$= 0.877(R' - Y)$$

$$R' = Y + 1.140V$$

$$G' = Y - 0.395U - 0.581V$$

$$B' = Y + 2.032U$$

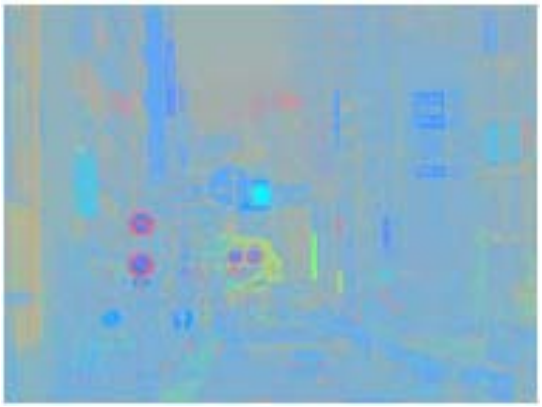


Normalized RGB

$$r = \frac{R}{R+G+B}$$
$$g = \frac{G}{R+G+B}$$
$$b = \frac{B}{R+G+B}$$

where $r+g+b=1$





Normalized RGB

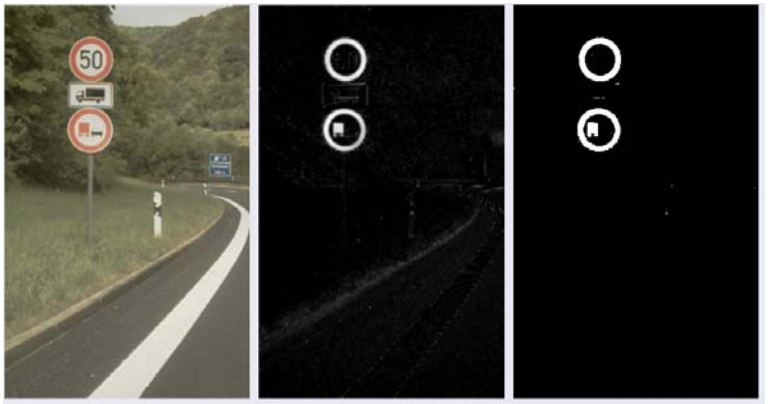




Ruta RGB

- Useful to find the pixels with dominant R/G/B color

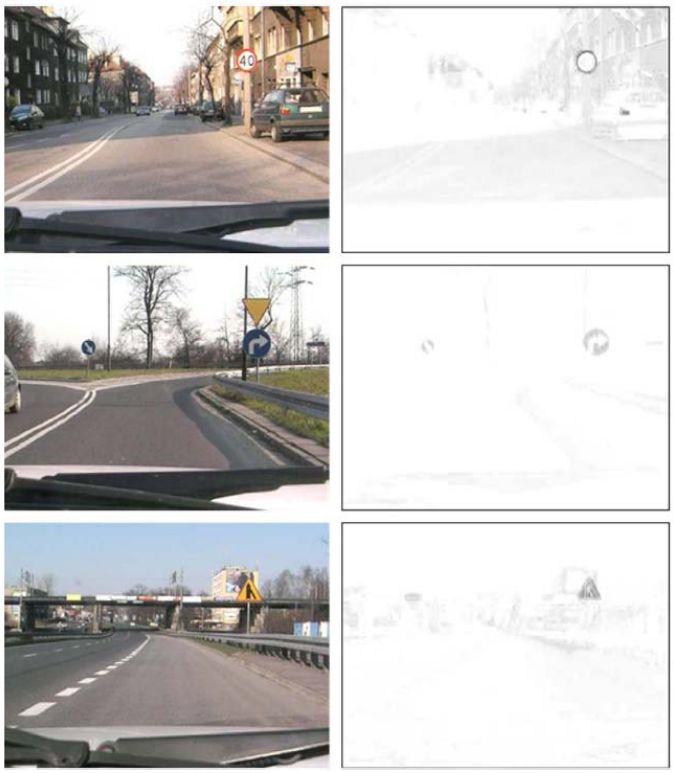
$$f_R = \max(0, \frac{\min(R-G, R-B)}{S})$$
$$f_G = \max(0, \frac{\min(G-R, G-B)}{S})$$
$$f_B = \max(0, \frac{\min(B-R, B-R)}{S})$$
$$f_Y = \max(0, \frac{\min(R-B, G-B)}{S})$$
where $S = R+G+B$



(a) Original (b) Enhanced [Ruta10] (c) Thresholded

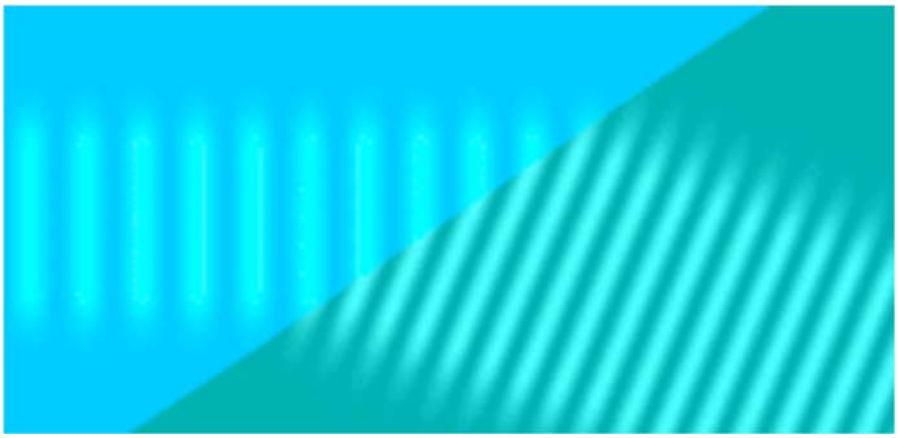
[1] A. Ruta, Y. Li, and X. Liu. Real-time traffic sign recognition from video by class-specific discriminative features. Pattern Recognition, 43(1):416–430, 2010.

Ruta RGB



[1] A. Ruta, Y. Li, and X. Liu. Real-time traffic sign recognition from video by class-specific discriminative features. Pattern Recognition, 43(1):416–430, 2010.

Thank You
Next : Optics



Computer Vision
Report 02: Color Transform

Yeungnam University
Dept. of Information and Communication
Ho-Youl JUNG

영남대학교 Yeungnam University

주요내용

- 주어진 RGB 컬러 영상을 YCbCr 로 변환하는 프로그램
- 변환된 YCbCr 을 다시 RGB 컬러 영상으로 변환하는 프로그램 작성

BGR2YCrCB

- 아래 주어진 수식 등을 통하여 RGB 영상을 YCbCr 영상으로 변형한다.

➢ For SDTV

$$Y_{601} = 0.299R' + 0.587G' + 0.114B'$$
$$Cb = -0.172R' - 0.339G' + 0.511B' + 128$$
$$Cr = 0.511R' - 0.428G' - 0.083B' + 128$$

➢ For HDTV

$$Y_{709} = 0.213R' + 0.715G' + 0.072B'$$
$$Cb = -0.117R' - 0.394G' + 0.511B' + 128$$
$$Cr = 0.511R' - 0.464G' - 0.047B' + 128$$

➢ For SDTV: computer system

$$Y_{601} = 0.257R' + 0.504G' + 0.098B' + 16$$
$$Cb = -0.148R' - 0.291G' + 0.439B' + 128$$
$$Cr = 0.439R' - 0.368G' - 0.071B' + 128$$

➢ For HDTV : Computer system

$$Y_{709} = 0.183R' + 0.614G' + 0.062B' + 16$$
$$Cb = -0.101R' - 0.338G' + 0.439B' + 128$$
$$Cr = 0.439R' - 0.399G' - 0.040B' + 128$$

YCrCB2BGR

- 아래 주어진 수식 등을 통하여 YCbCr 영상을 RGB 영상으로 변형한다.
- 아래 주어진 수식은 16~235범위의 RGB(각 8bit)를 YCbCr으로 변환하는 식이다.

➢ For SDTV

$$R' = Y_{601} + 1.371(Cr - 128)$$
$$G' = Y_{601} - 0.698(Cr - 128) - 0.336(Cb - 128)$$
$$B' = Y_{601} + 1.732(Cb - 128)$$

➢ For HDTV

$$R' = Y_{709} + 1.540(Cr - 128)$$
$$G' = Y_{709} - 0.459(Cr - 128) - 0.183(Cb - 128)$$
$$B' = Y_{709} + 1.816(Cb - 128)$$

➢ For SDTV: computer system

$$R' = 1.164(Y_{601} - 16) + 1.596(Cr - 128)$$
$$G' = 1.164(Y_{601} - 16) - 0.813(Cr - 128) - 0.391(Cb - 128)$$
$$B' = 1.164(Y_{601} - 16) + 2.018(Cb - 128)$$

➢ For HDTV : Computer system

$$R' = 1.164(Y_{709} - 16) + 1.793(Cr - 128)$$
$$G' = 1.164(Y_{709} - 16) - 0.534(Cr - 128) - 0.213(Cb - 128)$$
$$B' = 1.164(Y_{709} - 16) + 2.115(Cb - 128)$$

Report #2

- 주어진 수식을 통하여 구현한 프로그램의 코드와 실행 결과를 캡처하고 워드 또는 아래 한글에 붙여 넣어 아래 한글파일을 강의지원에 제출

- 파일명: report02_성명_학번.hwp

- 2가지 방법으로 구현 (RGB(각 8bits)➔YCbCr (각 8bits) ➔RGB (각 8bits)) 하고 그 결과를 비교
  - 구현1: OpenCV 함수 (cvtColor 함수) 를 이용하여 RGB를 YCbCr로 변환 후 Y채널만 추출해 흑백 영상으로 변환하고 이를 다시 RGB로 복원 구현
  - 구현 2: 수식을 이용하여 직접 구현
  - 구현 3: time 모듈을 사용하여 OpenCV 함수와 직접 구현한 함수의 실행시간 비교
  - Python 또는 C++ 중 하나만 선택하여 구현

Report #2 실행 결과 예)

- 구현 1 과 구현 2를 비교
- –원영상, Y영상, 복원된 RBG 영상을 차례로 디스플레이 함

